DAV PUBLIC SCHOOL, SECTOR-4 **Name:** _____

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Solutions to Exercise Set 11.1

Question 1:

Add two 4-digit numbers using the steps we have written down. Make sure you follow the steps precisely; do not perform any action that is not explicitly mentioned. Are you able to obtain the correct result?

Solution:

Yes, following the steps precisely yields the correct result. By systematically moving from the rightmost column (units) to the leftmost column (thousands), adding the two digits along with any carry from the previous step, and recording the units digit while passing the carry forward, the algorithm accurately completes the 4-digit addition.

Question 2:

What happens if you add a 5-digit number to a 3-digit number? Do our steps handle this situation correctly?

Solution:

If followed strictly as written, the algorithm **fails** because Step 3 explicitly specifies adding *"the two digits"* in the column. When a 5-digit number is added to a 3-digit number, the 4th and 5th columns from the right will only contain a single digit from the larger number.

Correction/Fix: To make the algorithm handle this correctly, we must pad the shorter (3-digit) number with **leading zeros** on the left to transform it into a 5-digit format (e.g., treating `345` as `00345`). This ensures every column has exactly two digits to add.

Question 3:

Why is it important to align the columns from right to left?

Solution:

Aligning numbers from right to left ensures that digits with the **same place value** are placed in the same column (units under units, tens under tens, hundreds under hundreds, etc.). If aligned incorrectly (e.g., from left to right), digits representing entirely different place values would be added together (such as adding hundreds to tens), leading to an mathematically incorrect sum.

Question 4:

In Step 3, why cannot the value of carry be more than 1?

Solution:

The maximum value of any single digit in base-10 is `9`. When adding a single column, the largest possible sum comes from adding the two largest digits (`9 + 9`) and the maximum possible incoming carry from the previous column (`1`):

```
Maximum Sum = 9 + 9 + 1 = 19
```

Since the highest possible sum is `19`, the value in the tens place (which becomes the carry for the next column) can **never be greater than 1**.

Question 5:

What happens if we do not include the fifth step in the algorithm above? Give examples where the algorithm will work correctly and where it will fail to work.

Solution:

Step 5 states: *"If the value of carry is 1, write 1 to the left of the bottom row."* If we omit this step, any carry generated from the final (leftmost) column will be ignored, resulting in an incorrect, truncated total sum whenever the answer requires an additional place-value digit.

Where it works correctly (No final carry generated):
When the sum of the leftmost digits plus any incoming carry is less than 10.
Example: `42 + 35 = 77`. In the tens column, `4 + 3 = 7` (carry is 0). Step 5 is not needed, so the answer is correct.

Where it fails (Final carry is ignored):
When the sum of the leftmost digits plus any incoming carry is 10 or greater.
Example: `75 + 43 = 118`. In the tens column, `7 + 4 = 11`. We write down `1` and carry over `1`. Without Step 5, that final carry is lost, and the recorded answer would incorrectly be `18` instead of `118`.